UNIT 1 · 9 hours

Real, Reactive, Apparent Power and Power Factor



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
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Learning objectives

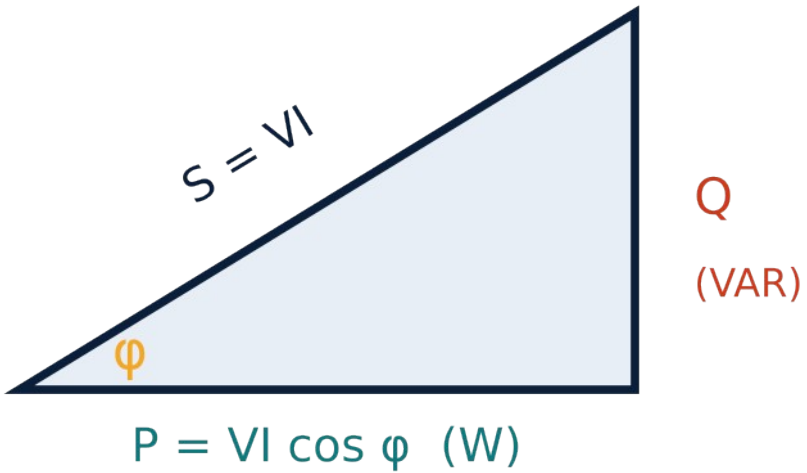
- Define P, Q and S and their units
- Draw and use the power triangle
- Define power factor, lagging and leading
- Explain why a low power factor is undesirable

Syllabus context

Analysis of R, L, C, RL, RC and RLC series circuits, real power, reactive power, apparent power and power factor.

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The power triangle



- Active (real) power P = VI cos φ (W)
- Reactive power Q = VI sin φ (VAR)
- Apparent power S = VI (VA)
- $S^2 = P^2 + Q^2$
- Power factor cos φ = P/S

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Meaning of each power

Active power P

Actual work done: heating, mechanical output. Measured by a wattmeter. Unit W or kW.

Reactive power Q

Power exchanged with L and C fields, no net work. Unit VAR. Q is positive for inductive loads.

Apparent power S

Product of RMS V and I. Rating of equipment: transformers and cables are rated in kVA.

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Power factor

Definition: cos φ = P/S = R/Z, where φ is the angle between V and I.

Lagging: inductive loads (motors, transformers); I lags V.

Leading: capacitive loads; I leads V.

Range: 0 to 1; unity for a pure resistor.

Low pf means the same real power needs a larger current.

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Why low power factor is bad

Consequences

- Larger current for the same power: $I = P/(V \cos \varphi)$
- More I^2R loss in cables and transformers
- Larger kVA rating of generators and lines
- Poor voltage regulation; utilities impose penalties

Improvement

- Connect capacitors in parallel with inductive loads
- Use synchronous condensers
- Avoid lightly loaded motors
- Target pf ≥ 0.95

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Numerical: power triangle

Problem A load takes 5 A at 230 V with a power factor of 0.8 lagging. Find S, P and Q.

- $S = VI = 230 \times 5 = 1150$ VA
- $P = S \cos\varphi = 1150 \times 0.8 = 920$ W
- $\sin\varphi = \sqrt{(1 - 0.8^2)} = 0.6$; $Q = S \sin\varphi = 690$ VAR

Answer: S = 1150 VA, P = 920 W, Q = 690 VAR

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Numerical: effect of power factor

Problem A 2 kW load on 230 V. Find the current at pf = 1 and at pf = 0.5.

- $I = P/(V \cos \varphi)$
- pf = 1: $I = 2000/230 = 8.70$ A
- pf = 0.5: $I = 2000/(230 \times 0.5) = 17.39$ A

Answer: Current doubles at pf = 0.5, so loss rises 4×

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Key takeaways

- $P = VI \cos \varphi$ (W), $Q = VI \sin \varphi$ (VAR), $S = VI$ (VA)
- $S^2 = P^2 + Q^2$; pf = $\cos \varphi = P/S$
- Low pf increases current, losses and equipment rating
- Capacitors in parallel improve a lagging pf

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Practice questions

Q1. A motor takes 10 A at 230 V, pf 0.7 lagging. Find P, Q and S.

Q2. Why are transformers rated in kVA and not kW?

Q3. Find the current of a 5 kW load at pf 0.8 on 230 V.

Q4. Draw the power triangle for a capacitive load.